Evaluating Functions from Rules, Tables, and Graphs
Answer Key

Grade 8 / Pre-Algebra

Quick Answers

1. 10	4. 6	7. 4	10. 28
2. 4	5. 5	8. —	11. —
3. 16	6. 2	9. 13	12. —

Curriculum

Level: Grade 8 / Pre-Algebra
8.F.A.1–8.F.B.5 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12*
Difficulty 1–5 of 5 across 12 tagged checks.

Common wrong answers

2. *If they got -14: squared only after applying the minus sign, computing -(3^2)-5*

1. $f(x) = 3x + 4$, so $f(2)$ replaces every x with 2: $f(2) = 3(2) + 4 = 6 + 4$. Multiply before adding. $\boxed{10}$

2. $f(x) = x^2 - 5$ with input -3. Substitute inside parentheses: $f(-3) = (-3)^2 - 5$. A negative squared is positive, so $(-3)^2 = 9$ and $9 - 5 = 4$. $\boxed{4}$
If a student writes $-3^2 - 5 = -14$, they squared the 3 and kept the sign outside; the parentheses are what prevent that.

3. Reading a table needs no arithmetic — find the input in the top row and read straight down. The column headed $x = 4$ has $h(x) = 16$ underneath, so $h(4) = \boxed{16}$.

4. $g(x) = \dfrac{x+7}{2}$ with input 5. Substitute first, then simplify the numerator before dividing: $g(5) = \dfrac{5+7}{2} = \dfrac{12}{2}$. $\boxed{6}$

5. Reading a graph: go to $x = 3$ on the horizontal axis, move up to the solid line, then across to the vertical axis. The solid line passes through the grid point $(3, 5)$, so $f(3) = \boxed{5}$. The rule $f(x) = 2x - 1$ confirms it: $2(3) - 1 = 5$.

6. The question asks for an *input*, not an output, so scan the bottom row for the smallest value and then read the input above it. The outputs are 8, 5, 4, 5, 8; the smallest is 4, and it sits above the input 2. $\boxed{2}$

7. Use the dashed line for g. At $x = 1$ the dashed line passes through the grid point $(1, 4)$, so $g(1) = \boxed{4}$. The rule $g(x) = -x + 5$ agrees: $-1 + 5 = 4$.

8. Evaluate each rule at the same input, then compare the two outputs. $f(1) = 2(1) - 1 = 1$ and $g(1) = -1 + 5 = 4$. Since 1 is less than 4, the solid line sits below the dashed line at that input: $\boxed{<}$

9. $f(x) = 5 - 2x$ with input -4. Substitute: $f(-4) = 5 - 2(-4)$. Subtracting a negative adds, so $5 + 8 = 13$. $\boxed{13}$

10. "How much more" is a difference of two outputs read from the same table. $C(6) = 47$ dollars and $C(2) = 19$ dollars, so the increase is $47 - 19$ dollars. (As a check, four extra months at 7 dollars per month is also 28.) $\boxed{28}$

11. Evaluate both rules at the input 4: $f(4) = 2(4) - 1 = 7$ and $g(4) = -4 + 5 = 1$. Since 7 is greater than 1, the solid line is above the dashed line there: $\boxed{>}$

12. Manual review — open explanation. A complete answer includes:

- Both rules are evaluated at the same input and produce the same output: $f(2) = 2(2) - 1 = 3$ and $g(2) = -2 + 5 = 3$.
- A point on a graph is an (input, output) pair, so a shared point means the two functions agree there — the crossing point is where $f(x) = g(x)$.
- To the left of that input f is below g (for example at the input 1, the outputs are 1 and 4); to the right f is above g (at the input 4, the outputs are 7 and 1). f grows as the input grows while g shrinks, so they can only cross once.

Accept any explanation that names the shared output at the shared input and uses at least one pair of values as evidence.